
BUSINESS ANALYSIS REPORT

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SECTION 1: CONDUCT OF ANALYSIS

1.1 Analysis Methods Used

The team employed the following data collection methods to gather information about (National Food Safety Institute) NFSI's existing processes and system requirements:

- Interview with Ms. Arinda Sophie (Director, Innovation & Digitization) on 24th March 2026
- Questionnaire administered to administrative staff
- Observation of the manual registration process

1.2 End-Users Contacted

The following end-users were identified and consulted during the business analysis process:

- Ms. Arinda Sophie – Director, Innovation and Digitization
- Lead Trainer or Training Coordinator

1.3 Records, Forms and Reports Analysed

The following existing records, forms and reports were reviewed to understand NFSI's current data management practices:

- Paper-based trainee registration forms
- Excel spreadsheets used for tracking payments
- WhatsApp messages used for client communication
- Manually issued certificates

1.4 Processes Observed

The following core business processes were observed and documented during the analysis phase:

- Trainee Registration Process; where a new trainee registers for a course (via email or WhatsApp)
 - Payment Collection Process; after payments are done, receipts are made manually.
 - Attendance Recording Process; attendance is recorded through roll calls.
 - Certificate Issuance Process; certificates are manually prepared and take 1-2 weeks to be delivered.
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1.5 Problems Encountered with Data Collection

- Limited staff availability for interviews due to active training schedules during analysis period
 - Some interviews had to be rescheduled; follow-up questions were handled via WhatsApp
- Difficulty quantifying exact volumes since no formal logs existed

1.6 Data Collection Instruments

The following instruments were developed and used to guide data collection:

- Interview Guide — A structured set of questions prepared for the interview.
- Observation Checklist — A checklist used during process observation to ensure all key workflow steps were documented including registration, payment, attendance and certificate issuance
- Document Review Checklist — A checklist used to systematically review existing records ensuring no critical document type was overlooked
- Questionnaire for Staff — A written questionnaire distributed to the administrative officer to capture daily transaction volumes and frequency of specific tasks

1.7 Synopsis of Interviews

Interview with Ms. Arinda Sophie — Director, Innovation & Digitization

Date	24th March 2026
Location	NFSI Offices, Ntinda, Kampala, Uganda
Duration	Approximately 90 minutes
Method	Structured face-to-face interview using questionnaire guide
Interviewers	Atwine Sheila, Nabachwa Priscilla, Naluyima Laureen, Kizito Juma, Mujuni Joel

Key findings from the interview:

- Ms. Arinda confirmed that NFSI currently manages all training operations manually using a combination of paper forms, Excel spreadsheets, email and WhatsApp communication
 - The Director highlighted delayed certificate issuance, lost trainee records and difficulty tracking payment status as the three most critical operational problems
 - She expressed strong enthusiasm for a digital Training Management Information System (TMIS) and confirmed that NFSI leadership fully supports the digitization initiative
 - The Director indicated that most NFSI clients already prefer virtual and online training confirming strong market alignment with a web-based platform
 - Staff adaptability was assessed as high; the Director estimated that all staff could adapt to the new system within less than one month with proper training and user manuals
 - Budget flexibility was indicated with the development cost estimate of UGX 22,000,000 considered acceptable given the anticipated benefits
 - The Director confirmed that no integration with existing external government or regulatory systems is required simplifying the technical scope
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SECTION 2: SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

2.1 External Interfaces

- User Interface; Web browser (Chrome, Firefox) and mobile app (Android/iOS)
- Payment interface; integration with MTN and Airtel Mobile Money
- Communication Interface; SMS gateway and email

2.2 System Functions

The NFSI TMIS must provide the following core functional capabilities:

- Trainee self-registration and profile management
- Course catalogue browsing and enrolment
- Automated certificate generation upon course completion
- Payment processing and invoice generation
- Attendance tracking per session
- Admin dashboard for monitoring registrations and payments
- Notification management

2.3 Usability Requirements

2.3.1 Learnability

- New staff must be able to perform basic tasks (registering a trainee, recording attendance)
- The system must provide a guided on boarding tutorial for first-time users on all platforms
- Help tooltips must be available on every form field to guide users through data entry
- A searchable help documentation section must be accessible from every page of the system

2.3.2 Accessibility

- The system must function adequately on internet connections of both low-end and high-end devices.
- Font sizes must be adjustable to accommodate users with visual impairments
- The system must remain functional on low-bandwidth connections (as low as 3G)
- The interface must be fully responsive adapting automatically to screen sizes from 4 inches (smartphones) to 15 inches (laptops)

2.3.3 Error Handling and Prevention

- The system must display clear, plain-language error messages (e.g. "Please enter a valid phone number" instead of "Error 404")
 - Forms must validate input in real-time before submission to prevent incorrect data entry
 - The system must request confirmation before any deletion action (e.g. 'Are you sure you want to delete this trainee record?')
 - Users must be able to undo accidental actions where possible
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2.3.4 Consistency and Efficiency

- All pages must follow a consistent layout and colour scheme aligned with NFSI's institutional branding
- Button labels, icons and terminology must remain uniform throughout the system (e.g. always use "Enroll" not sometimes "Register" and sometimes "Sign Up")
- Core tasks must be completable within a maximum of 5 steps from the main dashboard
- The system must support bulk enrollment of multiple trainees via CSV upload
- Admin staff must be able to generate a full attendance report in under 3 clicks

2.3.5 User Feedback

- The system must display immediate visual feedback after every user action (e.g. success message after registration submission)
- Loading indicators must appear whenever a process takes longer than 2 seconds
- SMS/email confirmation must be sent to trainees within 1 minute of successful registration
- Each user role must see a customized dashboard showing only what is relevant to their function

2.4 Performance Requirements

2.4.1 Response Time

Operation	Maximum Response Time
Page load (any screen)	3 seconds under normal network conditions
Search / trainee record retrieval	2 seconds
Certificate generation	10 seconds after triggering
Payment confirmation display	5 seconds after transaction
Dashboard report rendering	4 seconds
SMS/Email notification dispatch	Within 1 minute of triggering event
Standard report generation	5 seconds

2.4.2 Capacity and Throughput

- The system must support at least 100 simultaneous users without performance degradation
 - During peak enrollment periods the system must maintain normal performance with up to 200 concurrent users
 - The payment module must handle at least 50 concurrent payment transactions without failure
 - The certificate generation module must produce at least 60 certificates per hour
 - The database must store records for at least 10,000 trainees without degradation
 - The system must retain at least 5 years of training, payment and attendance records
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2.4.3 Availability and Reliability

- System uptime must be a minimum of 99% during core business hours (8:00 AM – 6:00 PM, Monday to Friday)
- Scheduled maintenance must only occur outside business hours (midnight to 4:00 AM)
- In the event of unexpected downtime, the system must be restored within 2 hours
- All payment transactions must either complete fully or roll back entirely

2.4.4 Scalability

- The system must scale from 80 registrations per month (current) to 500+ registrations per month without redesign
- Cloud hosting must allow both vertical and horizontal scaling as NFSI grows regionally
- Auto-scaling must be supported to provide additional resources during peak periods automatically

2.5 Logical Database Requirements

2.5.1 Core Data Entities

Entity	Key Attributes
Trainee	TraineeID, Full Name, Date of Birth, Gender, National ID, Phone, Email, Organisation, Address, Date Registered, Account Status
Course	CourseID, Course Name, Description, Category, Duration, Mode of Delivery, Max Capacity, Course Fee (UGX), Status, Prerequisites
Training Session	SessionID, CourseID, Session Date, Start/End Time, Venue/Platform, TrainerID, Session Status
Enrollment	EnrollmentID, TraineeID, CourseID, Enrollment Date, Enrollment Status, Payment Status
Attendance	AttendanceID, SessionID, TraineeID, Check-in Time, Check-out Time, Attendance Status, Recorded By
Payment	PaymentID, TraineeID, EnrollmentID, Amount Paid, Payment Date, Payment Method, Transaction Reference, Status, Invoice Number
Certificate	CertificateID, TraineeID, CourseID, Issue Date, Serial Number, Status, Issued By
Staff	StaffID, Full Name, Role, Email, Phone, Department, Account Status, Date Joined
Notification	NotificationID, RecipientID, Type (SMS/Email), Message Content, Date Sent, Delivery Status

2.5.2 Key Relationships

- One Trainee can enroll in many Courses (One-to-Many)
 - One Course can have many Training Sessions (One-to-Many)
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- One Training Session can have many Attendance records (One-to-Many)
 - One Enrollment must have one Payment record (One-to-One)
 - One Trainee can receive many Certificates across different courses (One-to-Many)
 - One Staff member can conduct many Training Sessions (One-to-Many)

2.5.3 Data Integrity Rules

- A Certificate cannot be issued unless the Trainee's Enrollment Status is 'Completed' and Payment Status is 'Paid'
- An Enrollment cannot be created for a Course that has reached its maximum capacity
- Duplicate trainee records must be prevented by validating National ID and email before creating a new profile
- Deleting a Trainee record must archive associated payment and certificate records; not permanently erase them
- All date fields must follow ISO 8601 format (YYYY-MM-DD) for consistency

2.5.4 Estimated Data Volumes

Entity	Monthly Records	Annual Records
Trainees (new registrations)	50–80	~700
Enrollments	50–80	~700
Training Sessions	10–15	~150
Attendance Records	500–1,000 entries	~10,000 entries
Payment Records	50–80	~700
Certificates Issued	40–60	~600
Notifications Sent	200–400 messages	~4,000 messages

2.5.5 Backup and Recovery

- Automatic daily backups must be performed at midnight and stored in a geographically separate cloud location
- Records older than 5 years must be automatically archived while remaining searchable
- In the event of data loss the system must be restorable to the last 24-hour backup within 1 hour

2.6 Design Constraints

2.6.1 Hardware Constraints

- The system must run on existing NFSI hardware (laptops and smartphones) without requiring new device purchases
 - All processing must be handled server-side (cloud) to reduce burden on client devices
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- The system must support printing of certificates and invoices through standard office printers already available at NFSI

2.6.2 Software Constraints

- The system must be compatible with different web browsers like Google Chrome, Mozilla Firefox
- The mobile application must support Android and iOS
- The system must not require installation of additional software beyond a standard web browser
- The backend must use open-source technologies to minimize licensing costs
- The system must support Microsoft Office formats (Excel, Word) for data import and export

2.6.3 Regulatory and Legal Constraints

- The system must comply with the Uganda Data Protection and Privacy Act 2019
- Trainee personal data must be collected only with explicit consent
- Payment processing must comply with Bank of Uganda electronic payment regulations
- The system must comply with ISO/IEC 27001 information security management standards

2.6.4 Budgetary Constraints

- Total development cost must not exceed UGX 22,000,000 as established in the Feasibility Study Report
- Technology choices must prioritize free and open-source software to avoid licensing fees
- Annual operating costs must not exceed UGX 3,100,000 as projected in the FSR

2.6.5 Security Constraints

- Role-based access control must ensure each user only accesses features relevant to their role
- All passwords must be stored using bcrypt hashing - plain text storage is strictly prohibited
- Multi-factor authentication must be enforced for all administrator accounts
- Session timeout must occur after 15 minutes of inactivity

2.7 Software System Attributes

2.7.1 Reliability

- Minimum uptime of 99% during core business hours (8:00 AM – 6:00 PM, Monday to Friday)
 - Mean Time Between Failures of at least 720 hours (30 days)
 - Automatic error detection and logging of system faults immediately upon occurrence
 - All payment transactions must be atomic; complete fully or roll back entirely
 - Data redundancy through cloud replication to eliminate single point of failure
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2.7.2 Security

- Role-based access control (RBAC) with clearly defined permissions per user role
- All data transmitted over HTTPS with TLS 1.2 or higher encryption
- Passwords stored using bcrypt hashing algorithm
- Complete tamper-proof audit trail recording every login, data change and administrative action
- IP blocking for suspicious repeated access attempts

2.7.3 Maintainability

- Modular architecture where each component can be updated independently without affecting others
- Version control implemented using Git with a clear branching strategy
- Dedicated staging environment for testing updates before applying to the live system

2.7.4 Portability

- Web application must function correctly across all major browsers without browser-specific modifications
- System must be deployable on multiple cloud hosting providers without significant code changes
- Database must be exportable and migrable to a different DBMS if required

2.7.5 Scalability

- Horizontal scaling; additional servers can be added to handle increased load
- Vertical scaling; storage and processing capacity can be increased without schema redesign
- Auto-scaling to automatically provide additional cloud resources during peak periods

2.7.6 Recoverability

- Full system recovery within 2 hours following a major failure or data loss event
 - One-click rollback to the previous stable version following a failed deployment
 - All incomplete transactions automatically detected and reversed to maintain data consistency
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SECTION 3: INFRASTRUCTURE ANALYSIS

3.1 Hardware

Device	Minimum Specification	Usage	Status
Laptops	2GB RAM, Intel Core i3 or equivalent	Admin, trainers, management access	Available
Smartphones	Android 8.0+, 2GB RAM	Mobile access for trainers and trainees	Available
Cloud Web Server	2 vCPUs, 4GB RAM, 80GB SSD	Hosting the web application	To be provisioned
Cloud Database Server	2 vCPUs, 4GB RAM, 100GB SSD	Storing all TMIS data	To be provisioned
Printers	Standard inkjet/laser	Printing certificates and invoices	Available

Since NFSI already has sufficient laptops and smartphones confirmed during the interview with Ms. Arinda Sophie, no new client-side hardware purchases are required for initial deployment.

3.2 Software

Software	Type	Purpose	License
Ubuntu Server 22.04 LTS	Operating System	Running the web and database servers	Free
React.js / Vue.js	Frontend	Web user interface	Free
Laravel (PHP 8.x)	Backend Framework	Building server-side application logic and APIs	Free
MySQL 8.0 / PostgreSQL	Database	Storing and managing all TMIS data	Free
Flutter	Mobile Framework	Android mobile application	Free
MTN Momo/ Airtel Money API	Payment Gateway	Processing mobile money payments	Transaction fee
Africa's Talking API	SMS Gateway	SMS notifications to trainees and staff	~UGX500/SMS
Let's Encrypt	Security	Encrypting all data transmission (HTTPS)	Free

3.3 Networking

3.3.1 Current Network Status at NFSI

NFSI currently has reliable broadband(120Mbps) internet available at their Ntinda, Kampala office confirmed during the interview with Ms. Arinda Sophie. Staff already use internet-dependent tools including Zoom, Google Meet, Microsoft Teams, email and WhatsApp, indicating sufficient bandwidth for a web-based system. The existing network infrastructure is adequate for the proposed TMIS without requiring major upgrades.

3.3.2 Recommended Network and Cloud Network Architecture

Component	Specification	Purpose
Cloud Load Balancer	Auto-distributes traffic	Handling peak enrollment periods without downtime
Backup Internet	4G LTE Mobile Router	Failover connectivity during primary internet outages
SSL/TLS (Cloud)	TLS 1.2+ on all endpoints	Encrypting all data between users and server
Firewall	Software or hardware firewall	Protecting the internal network from unauthorized access
VPN (Optional)	Virtual Private Network	Secure remote access for administrators working off-site

3.4 Standards

Standard	Category	Relevance to NFSI TMIS
Uganda Data Protection and Privacy Act 2019	Legal	All trainee personal data must be collected with consent and stored securely.
Bank of Uganda Electronic Payment Regulations	Legal	The mobile money payment module must comply with these regulations
ISO 22000 / Codex Alimentarius	Food Safety	NFSI training content and certificates must align with these standards

Standard	Category	Relevance to NFSI TMIS
ISO/EC 27001	Security	Guides the overall information security architecture of the system
OWASAP Top 10	Development	All code must address the top 10 web application security vulnerabilities
ISO 8601 / UTF-8 / PDF/A	Data	Standard format for dates, text encoding and certificate archiving.
Electronic Transactions Act 2011 (Uganda)	Legal	Digitally issued certificates are legally valid under this Act

CONCLUSION

This Business Analysis Report has provided comprehensive documentation of the analysis activities conducted for the proposed Training Management Information System (TMIS) for the National Food Safety Institute (NFSI).

The Conduct of Analysis section documented the data collection methods employed, the end-users consulted, the existing records and processes examined, and the challenges encountered during the analysis phase. The analysis confirmed that NFSI's current manual processes are a significant operational problem causing delays in registration, certificate issuance and payment tracking.

The Software Requirements Specification section formally defined what the proposed TMIS must do and how it must perform, covering external interfaces, system functions, usability requirements, performance requirements, logical database requirements, design constraints and software system attributes. These requirements are grounded in the realities of NFSI's operational environment and the organization's budgetary constraints.

The Infrastructure Analysis section assessed the hardware, software, networking and standards required to support the proposed system. The analysis confirmed that NFSI's existing hardware and internet infrastructure are adequate for the system with the primary recommendation being cloud-based hosting to eliminate the need for on-premise servers.
